

# Optimizing InGaN Micro-LED Efficiency: Investigating the Internal Quantum Efficiency and Ideality Factor Connection

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**Abstract**—We present a significant enhancement of internal quantum efficiency (IQE) exceeding 90% in InGaN-based Micro-light-emitting diodes (Micro-LEDs) achieved through sidewall passivation using atomic layer deposition (ALD). An intriguing observation is an inverse correlation between the diode ideality factor and IQE; devices boasting over 90% IQE consistently exhibit a diode ideality factor below 1.5. Specifically, for 10- $\mu\text{m}$ -sized devices, an IQE of 95.6% paired with a remarkably low ideality factor of 1.27 is realized, representing the pinnacle of achievable IQE at room temperature. In addition, enhanced light extraction efficiency (LEE) is observed through a low-refractive-index dielectric layer for passivation, facilitating a large critical angle at the interfaces. However, despite these advancements, the LEE for passivated devices remains at approximately 26%, resulting in an external quantum efficiency (EQE) of 25%. Consequently, our findings strongly advocate further device performance improvements in Micro-LEDs by enhancing LEE, necessitating novel device structures. In addition, we report the groundbreaking verification of the unity ideality factor for radiative recombination current in InGaN-based Micro-LEDs. This achievement is realized through the isolation of the radiative current component from the total current, thus substantiating the legitimacy of employing the Shockley–Read–Hall (SRH) to analyze the electrical characteristics of InGaN Micro-LEDs.

**Index Terms**—Ideality factor, internal quantum efficiency (IQE), micro-light-emitting diode (Micro-LED).

## I. INTRODUCTION

A NEW era of innovative displays is rapidly approaching, promising groundbreaking innovations such as folding

and rollable displays capable of projecting 3-D content and enabling extended reality (XR) experiences for the emerging metaverse. Several crucial attributes are essential to meet the diverse demands of this next-generation display, including wearable/flexible displays, XR or 3-D displays, and high-resolution (4/8/16K) large-screen TVs. These include high pixel density, high power efficiency, high luminance, fast refresh rate, and high environmental stability.

Micro-light-emitting diodes (Micro-LEDs) based on III-nitrides, with sizes  $\leq 100\ \mu\text{m}$ , have emerged as the leading technology to fulfill these demanding requirements. Their exceptional properties, such as self-illumination, high brightness, low power consumption, fast response time, and extended lifetime, make them highly promising [1], [2]. Micro-LEDs can achieve extremely high brightness ( $> 10^7\ \text{cd/m}^2$ ) [3], [4], subnanosecond response time [5], [6], high pixel density (exceeding 10k pixels per inch, PPI) [7], [8], [9], exceptional stability, and a broad operating temperature range [10], [11], [12].

The superior efficiency of Micro-LEDs can be attributed to various factors, including enhanced thermal distribution [13] and optimal current density distribution [14], lower junction temperature [15], and, most notably, the significant mitigation of sidewall damage [16] that typically occurs during their fabrication process. Suppressing sidewall damage is pivotal in enhancing Micro-LED efficiency by reducing nonradiative recombination centers at the surface of the active layers. These nonradiative recombination centers are often generated by defects or trap states at the sidewall surface, reducing radiative recombination in the active regions. As the size of devices shrinks, the quantum efficiency tends to decrease due to the larger volume-to-area ratio inherent to smaller devices [17].

Several techniques have been proposed to mitigate sidewall damage in Micro-LEDs, including wet chemical treatments [18], neutral beam pixelation [19], and passivation with dielectric layers using atomic layer deposition (ALD) [20] or plasma-enhanced chemical vapor deposition (PECVD) [21] to minimize dangling bonds on the sidewalls. Among these approaches, passivation is the most effective means of mitigating sidewall damage in Micro-LEDs. While PECVD offers a rapid deposition rate and reduces surface-related leakage current, it struggles with nonuniform sidewall coverage, leading to persistent current leakages from the sidewall of active layers. Conversely, ALD relies on surface chemical reactions, producing high-quality film deposition with excellent step coverage

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while preventing carrier trapping by surface defects [20]. Consequently, sidewall passivation achieved through the ALD technique has proven far more effective in enhancing quantum efficiency than PECVD [20], [22], [23].

Quantum efficiencies, including external quantum efficiency (EQE), internal quantum efficiency (IQE), and the diode ideality factor ( $n_d$ ), constitute critical parameters for evaluating the performance of LED devices. EQE quantifies the proportion of photons emitted from the device relative to the injected carriers. At the same time, IQE measures the ratio of emitted photons from the LED to the number of electron–hole pairs generated within the LED itself.

The diode ideality factor is a distinctive parameter that characterizes how closely a diode aligns with the ideal diode equation. It can be derived from the slope of the semilogarithmic plot of the  $I$ – $V$  curve. The ideality factor serves as a valuable indicator of carrier transport and recombination mechanisms [24] and offers insights into the degree of recombination in the active region of the LED. A lower ideality factor signifies that the LED effectively transports carriers across the junction and promotes radiative recombination.

In the context of classic models such as ABC and Shockley–Read–Hall (SRH) [24], [25],  $n_d = 1$  corresponds to direct band-to-band recombination, 2 indicates defect-associated SRH recombination, and 2/3 signifies Auger recombination [26]. Thus, the ideality factor of Micro-LEDs typically falls between 1 and 2 depending on their specific recombination nature.

While numerous reports have explored nonradiative recombination processes and associated ideality factors for the III-nitride-based LEDs, often revealing ideality factors greater than 2 [27], [28], [29], this phenomenon has been attributed to the potential tunneling leakage facilitated by deep defects [30], [31]. In III-nitride Micro-LEDs, the nonradiative recombination mechanisms under forward bias are not readily explained by SRH recombination via defect levels. Consequently, two fundamental questions emerge concerning III-nitride Micro-LEDs: 1) what is the relationship between quantum efficiencies and ideality factor and 2) is the ideality factor of the radiative recombination current indeed equal to 1?

The practical evaluation of device performance in Micro-LEDs hinges significantly on EQE, which stands as a critical product of IQE and the light extraction efficiency (LEE), denoted as  $\text{EQE} = \text{IQE} \times \text{LEE}$ . Numerous studies assessing the performance of Micro-LED devices have employed EQE as a primary metric for quantifying enhancements in performance. Specifically, Micro-LEDs equipped with dielectric passivation layers have shown size-independent EQE in the size 10–100- $\mu\text{m}$  range, often attaining a peak EQE of approximately 30% [20]. This trend implies improved IQE, assuming that LEE remains constant throughout [32]. However, while EQE has enjoyed extensive attention in the literature, IQE, a pivotal component of this equation, has been curiously underrepresented and insufficiently explored in the realm of Micro-LED research [4], [33].

The studies on the temperature dependence of the ideality factor in conventional InGaN-based LEDs (sizes of  $>200 \mu\text{m}$ ) shed light on its impact on IQE [34]. It leads to a hypothesis

that a higher IQE signifies more ideal diode behavior, resulting in a lower ideality factor. Anticipation, thus, builds for a robust correlation between IQE and the ideality factor, specifically in Micro-LED [26]. This fundamental question remains unanswered mainly within the Micro-LED community, leaving a significant knowledge gap.

Furthermore, the intricate relationship between IQE and Micro-LED's ideality factor has yet to be systematically scrutinized. Although the unity ideality factor of radiative recombination current has been reported for commercial InGaN/GaN and GaN p-n junction diodes [35], [36], an absence of such reports persists concerning Micro-LEDs. Hence, a pressing need exists to confirm this unity ideality factor, as doing so holds the promise of advancing our comprehension of the recombination mechanisms governing the active regions within Micro-LEDs.

In this study, we harnessed the potential of Micro-LEDs integrated with dielectric layer passivation to tackle the fundamental questions posed earlier. Our findings have illuminated a robust correlation between heightened IQE and the attainment of a lowered ideality factor. With the successful confirmation of the ideality factor's unity with Micro-LEDs, achieved through the isolation of the radiative recombination current, we have significantly strengthened the foundation for employing the SRH model in the electrical characterization of these devices. Consequently, our findings offer new insights and pave the way for future advancements in Micro-LED technology, carting a promising course for further device enhancements.

## II. MATERIALS AND METHODS

We successfully fabricated green InGaN-based Micro-LEDs emitting at a wavelength of 520 nm with varying dimensions: 10, 30, 50, 80, and 100  $\mu\text{m}$  on each side. These Micro-LEDs were manufactured on a commercially available epi wafer (Changelight Company Ltd.), which was grown on a sapphire substrate using the metal-organic chemical vapor deposition (MOCVD) method. The epitaxial structure of the c-plane GaN-based Micro-LEDs features a 523-nm emission wavelength and is composed of several layers: a 10–40-nm buffer layer, a 3000–4000-nm unintentionally doped GaN (u-GaN) layer, a 1500–3000-nm Si-doped n-type GaN (n-GaN) layer, a 250–400-nm multiquantum well (MQW) active region with ten periods, and a 100–300-nm Mg-doped p-type GaN (p-GaN) layer on the top.

The fabrication process involves several critical steps for both LED-A and LED-B. Initially, a 500-nm  $\text{SiO}_2$  layer is deposited using PECVD, followed by a pattern definition using photolithography. Sequential etching steps define the hard mask and the mesa. A current spreading layer (CSL) was created on the p-GaN surface using indium tin oxide (ITO). To establish ohmic contact between p-GaN and the CSL, rapid thermal annealing (RTA) was applied. Specifically, the annealing process involves heating at 550  $^\circ\text{C}$  under  $\text{O}_2$  conditions for 5 min, followed by 600  $^\circ\text{C}$  under  $\text{O}_2$  and  $\text{N}_2$  conditions for 3 min.

For LED-A, electrodes are evaporated directly after the ITO layer is finished. In contrast, LED-B undergoes additional

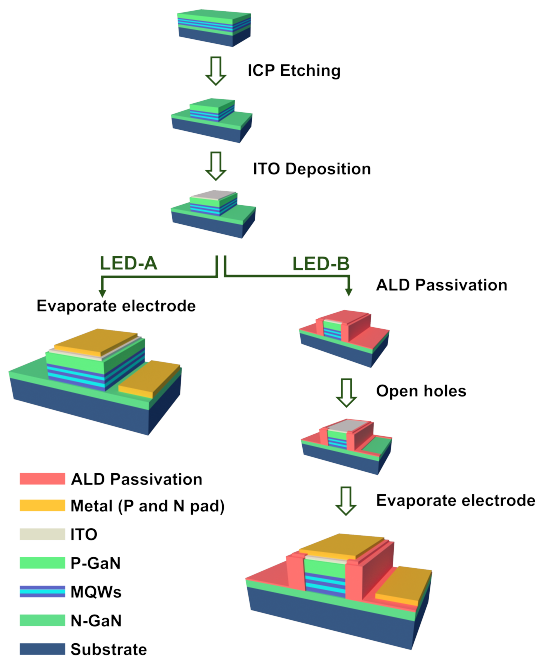


Fig. 1. Device fabrication process flow.

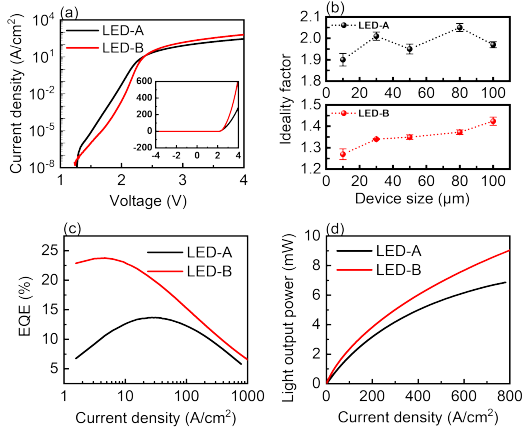


Fig. 2. (a) Logarithmic scale reveals the current density–voltage relationship for the 80- $\mu\text{m}$  device, while the inset displays the same relationship in a linear scale. (b) Ideality factor is derived from the  $J$ – $V$  curve across all device sizes. (c) EQEs for the 80- $\mu\text{m}$  devices. (d) Light output power for the 80- $\mu\text{m}$  device.

steps, including KOH immersion to remove residual debris, followed by thermal ALD at 300 °C for a 50-nm  $\text{Al}_2\text{O}_3$  passivation layer. Using photoresist as a mask, buffered oxide etch (BOE) wet etching opens holes for electrode placement, ensuring that the ALD layer remains only on the sidewalls and does not interfere with device performance. For the p and n electrodes, we utilized a layered structure consisting of Ti/Al/Ti/Au, with a thickness of 20/100/20/50 nm, deposited using electron beam evaporations (e-beam). The resultant devices, LED-A and LED-B, are illustrated in Fig. 1.

### III. RESULT AND DISCUSSION

Current and voltage measurements were conducted using Keysight (formerly Agilent) B1500 Analyzer. Fig. 2(a) displays the measured current density–voltage ( $J$ – $V$ ) curves

for 80- $\mu\text{m}$  devices. The impact of leakage current on LED  $J$ – $V$  characteristics is evident through nonzero currents under reverse bias or nonzero shunt currents at voltages lower than the turn-on voltage [37].

In the voltage range of 1.5–2.4 V, LED-A exhibits higher current densities, attributed to increased nonradiative recombination, resulting in higher shunt currents due to leakage paths created by sidewall damage during etching. The absence of a passivation layer leaves the device more susceptible to sidewall damage caused by etching. This damage can introduce additional leakage channels, increasing leakage current, especially at lower voltages. On the other hand, LED-B benefits from an  $\text{Al}_2\text{O}_3$  passivation layer, which effectively mitigates sidewall damage and reduces the number of leakage channels. This improvement in structural integrity leads to lower overall resistance in LED-B compared to LED-A, especially notable at lower voltages where leakage currents are more influential.

We determined the ideality factor using  $n_d = (qV/k_B T)((d(\ln J))/dV)^{-1}$ , where  $J$  is the forward current density,  $q$  is the electronic charge,  $V$  is the applied forward bias,  $n_d$  is the ideality factor,  $k_B$  is Boltzmann's constant, and  $T$  is the temperature. The ideality factor is notably high in the low current range due to the shunt resistance ( $R_{\text{shunt}}$ ) and tends to decrease as the current increases. Conversely, in the high current range, the series resistance ( $R_{\text{series}}$ ) becomes predominant, leading to a rise in the ideality factor with increasing current [29]. We have focused our analysis on the intermediate current density range, specifically between 0.1 and 1  $\text{A}/\text{cm}^2$ , where the space charge region predominantly influences the device's  $J$ – $V$  characteristics. This range is critical because the  $J$ – $V$  characteristic typically follows an exponential relation, presenting a local minimum value of the ideality factor. To determine the most relevant operating point for calculating the ideality factor, we have utilized the linear region of the  $V$  versus  $\ln J$  plot shown in Fig. 2(a), corresponding to the minimum value of  $n_{\text{ideality}}$ . This approach ensures that the calculated ideality factor reflects the device's behavior in a regime where the space charge region dominates. We have refined our calculation by considering the minimum value and the average of ten nearby points within this linear region, thereby establishing an effective ideality factor.

In Fig. 2(b), the ideality factor for the 80- $\mu\text{m}$  LED-B device ( $n_d = 1.37$ ) is significantly lower than that of the 80- $\mu\text{m}$  LED-A device ( $n_d = 2.05$ ). Under the p-n junction,  $n_d = 1$  is associated with band-to-band radiative recombination, and  $n_d = 2$  indicates SRH recombination via defect levels [24].  $n_d > 2$  is attributed to deep-level-assisted tunneling conduction [27], [28], [38]. Thus, the ideality factor of 1.4 suggests less nonradiative recombination for LED-B, possibly indicating deep-level defects at the sidewall surface caused by etching in LED-A.

Beyond 2.4 V, the current curve bends due to the influence of the series resistance. The overall resistance of LED-A is affected, with an observed increase in series resistance at higher operating voltages due to these leakage paths. The passivated device exhibits higher current due to lower resistance.



TABLE I  
PEAK EQE, IQE VALUES, AND LEE FOR THE 80- $\mu\text{m}$  DEVICE

|             | J at the peak<br>EQE ( $\text{A}/\text{cm}^2$ ) | EQE   | IQE   | LEE   |
|-------------|-------------------------------------------------|-------|-------|-------|
| LED-A       | 28.1                                            | 13.7% | 78.2% | 17.5% |
| LED-B       | 4.69                                            | 23.8% | 92.5% | 25.7% |
| Improvement |                                                 | 73.7% | 18.4% | 46.8% |

The EQEs of the 80- $\mu\text{m}$  LED-A and LED-B devices were measured by a commercialized system (XPQY-EQE-Adv, Guangzhou Xi Pu Optoelectronics Technology Company Ltd.) that was equipped with an integrated sphere and a photodetector array. Fig. 2(c) illustrates the EQE measurements for both devices. LED-B exhibits significantly higher EQE throughout the entire range of current densities. In addition, the current density at the peak EQE for LED-B is substantially lower. Specifically, the peak EQE of LED-A is 13.7% at 28.1  $\text{A}/\text{cm}^2$ , while LED-B reaches as high as 23.8% at 4.7  $\text{A}/\text{cm}^2$ . These results unequivocally underscore the remarkable impact of passivation, leading to an approximately 73.7% enhancement in peak EQE and a noteworthy reduction in the current density at the peak EQE.

In the ABC model, the current density at peak EQE ( $J_{\text{peak}}$ ) can be approximated as  $AB/C$ , [39] where A, B, and C represent the SRH, radiative, and Auger recombination rates, respectively. Since passivation primarily targets suppressing the defects at the sidewall surface, the values of B and C do not exhibit significant differences between passivated and nonpassivated devices at low current densities. Hence, near the peak EQE,  $J_{\text{peak}}$  is found to be directly proportional to A, the SRH nonradiative recombination rate. The starkly lower current density at the peak EQE for LED-B indicates reduced nonradiative recombination. Furthermore, the measured light output power of both Micro-LEDs, as shown in Fig. 2(d), confirms that LED-B with a passivation layer outperforms LED-A across all current densities.

The enhancement in IQE and LEE contributes to an overall increase in EQE. To dissect these contributions to EQE, we utilized the room-temperature reference-point method (RTRM) to obtain IQE [40], [41], while LEE was determined using EQE and the calculated IQE. The supporting material (IQE calculation part) reflects the calculation details. The summarized values are presented in Table I. Remarkably, the peak IQE/LEE values for LED-A and LED-B are 78.15%/17.49% and 92.51%/25.68%, respectively. Our results reveal that passivation leads to a 92.5% IQE, strongly indicating that the  $\text{Al}_2\text{O}_3$  passivation layer significantly suppresses defects at the sidewall of the active layer caused by ICP etching, a phenomenon well-documented in prior research [20], [21], [42].

Moreover, LEE improved approximately 47%, despite remaining relatively low. The increased LEE in passivated devices can be attributed to enhanced light extraction, a key advantage of dielectric sidewall passivation. According to Snell's law,  $\theta_{\text{crit}} = \sin^{-1}(n_1/n_2)$ , where  $\theta_{\text{crit}}$  is the critical angle,  $n_1$  represents the refractive index of air, and  $n_2$  is the

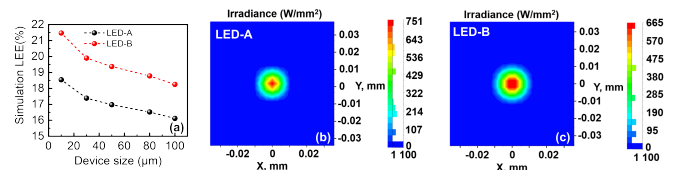


Fig. 3. Optical simulation results. (a) ALD passivation and without passivation for difference size devices' LEE simulation results. (b) Irradiance simulation results for the 10- $\mu\text{m}$  without passivation device. (c) Irradiance simulation results for the 10- $\mu\text{m}$  ALD passivation device.

refractive index of III-nitrides [20]. The critical angle for total internal reflection depends on the refractive index ratio, which is typically higher for most dielectric materials than for air. We increase light extraction by allowing a larger critical angle by depositing a dielectric layer with a lower refractive index than III-nitride.

To explore the effect of passivation on LEE, we performed Monte Carlo ray tracing simulations for different sizes of devices, both nonpassivated and ALD passivated. Monte Carlo ray tracing simulations were conducted using LightTools to investigate the impact of sidewall passivation on our Micro-LED device structures. The simulated model comprised a cubic geometry with each side measuring 10, 30, 50, 80, and 100  $\mu\text{m}$ . The device stack included a 2- $\mu\text{m}$ -thick sapphire substrate, a 2- $\mu\text{m}$ -thick n-type GaN (N-GaN) layer, a 0.125- $\mu\text{m}$ -thick MQW layer, and a 1- $\mu\text{m}$ -thick p-type GaN (P-GaN) layer. An additional 100-nm-thick  $\text{Al}_2\text{O}_3$  passivation layer was included for the passivated LED-B variant to encapsulate the sidewalls. The assigned refractive indices were 1.78 for the sapphire substrate, 2.5 for both N-GaN and P-GaN, 2.6 for the MQWs, and 1.77 for the  $\text{Al}_2\text{O}_3$  passivation layer. All contact surfaces within the model were interconnected using a glue interface to mimic real-world assembly.

In the simulation, light emission was considered from only one side of the device, necessitating the placement of a surface light source at the center of the MQW layer. The light source was set to emit within an angular range of  $0^\circ$ – $90^\circ$ , simulating a Lambertian distribution characteristic of the light emitted by Micro-LEDs. The central wavelength for the light source was set at 520 nm, with a maximum full-width at half-maximum (FWHM) of 30 nm.

A receiver slightly larger than the Micro-LED size (100  $\mu\text{m}$ ) was positioned above the model to capture the emitted light. Forward simulation ray tracing was then initiated to analyze the light distribution and quantify the effects of sidewall passivation.

Fig. 3(a) compares passivated and nonpassivated LEEs of different sizes. The LEE of ALD passivated devices is significantly improved, and as the size decreases, the LEE increases. This is because in smaller devices, lights are generally much closer to the sidewall, and more sidewall reflections occur, leading to higher LEE for smaller devices.

Since LEE has a more significant impact on small sizes, we show the most obvious contrast in the optical irradiance of the smallest size device (10  $\mu\text{m}$ ), as shown in Fig. 3(b) and (c). The optical irradiance of the passivated device surpasses that



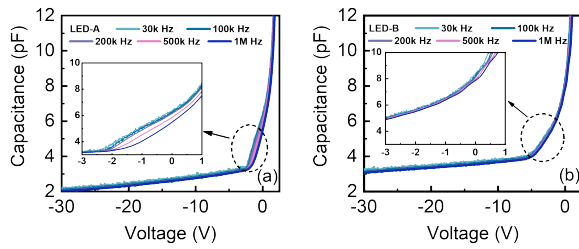


Fig. 4.  $C$ - $V$  measurement results for an 80- $\mu\text{m}$  device at different frequencies. (a) LED-A. (b) LED-B.

of the nonpassivated device. Specifically, dielectric passivation enhances LEE from 18.4% to 21.1% for a 10- $\mu\text{m}$  device, while LEE from 16.5% to 18.8% for an 80- $\mu\text{m}$  device, thereby confirming the positive impact of the dielectric passivation layer on LEE enhancement. Consequently, the improved EQE of the passivated device stems from enhanced IQE due to the suppression of defects at sidewall surfaces and improved LEE.

Additional evidence for the suppression of nonradiative recombination in LED-B comes from capacitance-voltage ( $C$ - $V$ ) measurements performed at various frequencies using Keysight (formerly Agilent) B1500 Analyzer. Fig. 4(a) and (b) displays modulation-frequency-dependent  $C$ - $V$  curves measured at room temperature for 80- $\mu\text{m}$  LED-A and LED-B, respectively.

In these measurements, capacitance significantly decreases within 2.5 to  $-2$  V for LED-A and 2.5 to  $-5$  V for LED-B. Subsequently, it exhibits a slower decrease within the voltage range from  $-2$  to  $-30$  V for LED-A and  $-5$  to  $-30$  V for LED-B. The change in slope indicates that the net fixed charge concentration in the depletion region is nonuniform, primarily due to differences in carrier injection concentration in devices [43]. LED-A experiences slope changes at  $-2$  V, while LED-B's slope changes at  $-5$  V. These later slope changes suggest that LED-B has more carrier injection or less carrier leakage, resulting from sidewall passivation protection.

Furthermore, as shown in the inset of Fig. 4, LED-A exhibits a noticeable drop in capacitance at higher frequencies, whereas the capacitance of LED-B remains unchanged with varying frequencies. The  $C$ - $V$  data of other devices of different sizes show similar trends. These observations can be attributed to deep traps, which trap carriers and hinder their mobility. Both fast free and slow trapped carriers can respond at low frequencies. However, the slow trapped carriers cannot respond quickly as the frequency increases, leading to decreased capacitance. Therefore, the frequency-dependent  $C$ - $V$  measurements reveal the presence of deep traps [44].

The results in Fig. 4 indicate that LED-A has deeper trap states, likely originating from sidewall damage caused by ICP etching. An increase in nonradiative recombination, associated with deep traps, reduces the IQE of the device and consequently affects the EQE. Therefore, LED-B achieves a higher IQE, resulting in improved EQE.

In Fig. 2(b), it is evident that nonpassivated devices exhibit higher ideality factors compared to their passivated counterparts. Furthermore, as the device size decreases, the ideality factor of passivated devices shows a noticeable decrease.

This trend aligns with the Shockley model, where a transition from an ideality factor of 2–1 signifies a reduction in nonradiative recombination, leading to the dominance of radiative processes. This explanation supports the ABC model. Auger recombination primarily impacts the EQE peak drop at a high injection current. Still, the discussion here focuses on ideality factors near turn-on voltage (typically less than 2.3 V), where injection currents are relatively low. Consequently, the Auger recombination coefficient  $C$  can be disregarded, simplifying IQE:

$$\frac{1}{\text{IQE}} \sim \frac{A}{\text{BN}} \quad (1)$$

It is evident from this equation that as coefficient  $A$  decreases, IQE rises. IQE signifies the ability of the LED's active region to convert injected electrical energy into optical energy, essentially representing the efficiency of radiative recombination. Conversely, the ideality factor reflects the mechanism governing current transport within the device.

The findings regarding IQE, EQE, and the ideality factor strongly indicate that a lower ideality factor correlates with higher IQE or EQE. Consequently, our results prompt us to explore the relationship between the ideality factor and IQE or EQE, explicitly focusing on size dependence. It is worth noting that no size dependence is observed in the ideality factor for nonpassivated devices, leading us to concentrate our study on passivated devices. We measured devices at five different locations and calculated the ideality factor to mitigate the potential influence of epitaxial defect, thus minimizing data randomness.

Fig. 5(a) illustrates the measured EQEs for passivated devices of varying sizes. As evident in Fig. 5(c), all devices exhibit peak EQEs exceeding 20%, indicating strong device performance due to the effective suppression of sidewall defects through passivation. Interestingly, these EQE curves demonstrate a mild size dependence, where EQE increases as device size decreases. Utilizing the EQE data, IQE was computed using the RTRM method [40], [41].

The size-dependent trend in the ideality factor and IQE, as shown in Fig. 5(b), reveals a consistent pattern: the ideality factor decreases. At the same time, IQE increases as the device size decreases. This phenomenon of increased IQE in smaller devices can be attributed to improved current spreading and enhanced heat dissipation capabilities. Smaller devices enable more balanced heat distribution and slower junction temperature rise [45], [46].

Similarly, the relationship between the ideality factor and EQE mirrors that of IQE and the ideality factor, as depicted in Fig. 5(c). However, it is challenging to establish a direct correlation between EQE and the ideality factor because LEE also influences EQE. Recent studies have highlighted that a longer peripheral length of the device can enhance LEE while reducing peak IQE [39]. Conversely, treatments such as tetramethylammonium hydroxide (TMAH) can mitigate etching damage and smoothen sidewall surfaces, increasing IQE. However, this can result in decreased EQE due to reduced LEE [47]. Therefore, the size dependence of EQE can vary depending on the LEE characteristics of the devices.

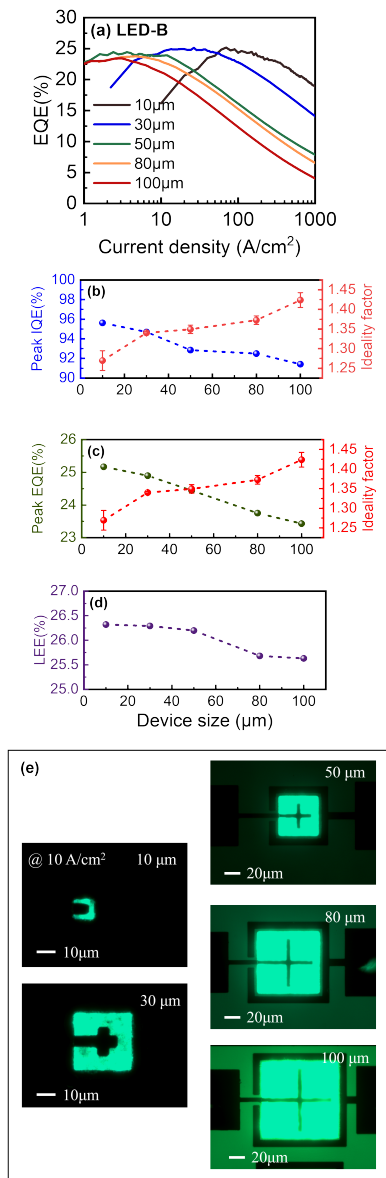


Fig. 5. (a) EQE measurements were taken for all sizes of LED-B. (b) Ideality factor and peak IQE for LED-B. (c) Ideality factor and peak EQE for LED-B. (d) LEE for LED-B. (e) Luminous images of different sizes.

The extracted LEE in Fig. 5(d) demonstrates very subtle size dependence. In other words, as the device size decreases, LEE tends to increase. This is primarily attributed to the fact that in smaller devices, light emission occurs closer to the sidewalls, resulting in more sidewall reflections and, consequently, higher LEE.

Fig. 5(e) shows the luminous images of different sizes. Since the emission from small-sized devices is weak at low current densities, to ensure consistent observation, images of various sizes were captured at 10 A/cm<sup>2</sup>. The 50-, 80-, and 100-μm devices were observed using the same magnification microscope, while the 10- and 30-μm devices were observed using a higher magnification microscope, resulting in inconsistent scales.

Our results show an ideality factor of 1.27 and an impressive 95.62% IQE for the 10-μm device, representing remarkable

achievements in Micro-LEDs based on our current knowledge. Notably, an ideality factor of 1.27 suggests the potential for achieving an ideality factor of 1, akin to an ideal diode in the SRH analysis [24].

SRH analysis has traditionally been employed to analyze the  $I$ - $V$  characteristics of curves of InGaN-based Micro-LEDs [27], [28], [36], [38]. However, it is worth noting that the SRH analysis was initially developed for Si diodes, which are indirect band gap material, where light emission through recombination was unlikely. In contrast, InGaN-based LEDs employ direct band gap materials, making radiative recombination the primary mechanism for light emission. Moreover, most InGaN-based Micro-LEDs are based on carrier confinement heterostructures, often incorporating multiple quantum well structures. It has been recognized that applying the SRH analysis to InGaN-based Micro-LEDs can be somewhat awkward though this aspect has received relatively little attention. As a result, high ideality factors exceeding 2 in InGaN-based Micro-LEDs have been attributed to modified SRH analysis that incorporates various other mechanisms [27], [28], [38]. Thus, verifying the unity ideality factor associated with radiative recombination current in InGaN-based Micro-LEDs becomes crucial, as it provides clear evidence for the validation of utilizing SRH analysis for InGaN-based Micro-LEDs.

The methods for separating and extracting radiative and nonradiative recombination currents are described in detail by Lee et al. [36]. To explain it briefly, the current density due to the radiative recombination,  $J_r$ , is obtained as  $J_r = \eta_{\text{int}}(J) * J$ , where  $\eta_{\text{int}}(J)$  is the IQE as a function of current density and  $J$  is the current density. The radiative current densities are a function of junction voltage  $V_j = V - R_s J_r A$ , where  $R_s$  is the series resistance and  $A$  is the device area. The  $R_s$  values for 80-μm LED-A and LED-B are 14.4 and 1.0 Ω, respectively.

The term “unity ideality factor” signifies that all recombinations within the LED are exclusively radiative recombinations. Such near-unity ideality factors have been reported for GaN p-n homojunction diodes [35]. Ideality factors exceeding 1, on the other hand, clearly indicate that the recombination process within the LED involves a combination of band-to-band radiative and nonradiative recombination. Separating these recombination processes within  $J$ - $V$  curves is important to know the contributions of radiative and nonradiative recombination to the ideality factor. Consequently, the ideality factor associated with radiative recombination in  $J$ - $V$  curves should ideally equal one, as demonstrated in conventional blue InGaN LED [36]. Surprisingly, no reports have confirmed the unity ideality factor of the radiative recombination currents for Micro-LEDs of any color. We extracted the radiative recombination currents from the measured  $J$ - $V$  curves for all devices to address this gap and determined the corresponding ideality factors.

Fig. 6(a) presents the calculated radiative recombination current as a function of junction voltage. The radiation recombination current of passivated devices is much larger than nonpassivated devices. Ideality factors for nonpassivated and passivated devices with a size of 80 μm are shown in Fig. 6(b) and (c). The ideality factor linked to the radiative recombination current for nonpassivated devices remains

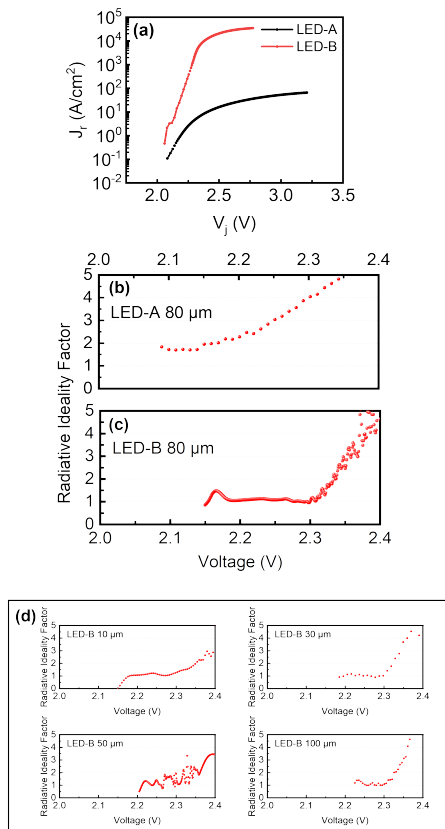


Fig. 6. (a) Radiative recombination current as a function of junction voltage. (b) Radiative ideality factor extracted from the LED-A 80- $\mu\text{m}$  device. (c) Radiative ideality factor extracted from the LED-B 80- $\mu\text{m}$  device. (d) Radiative ideality factor extracted from the LED-B other size device.

approximately 1.7, as depicted in Fig. 6(b). The unity ideality factor associated with the radiative recombination current is confirmed for the passivated device, as shown in Fig. 6(c). Also, the unity ideality factors are consistently achieved across passivation devices of various sizes, as shown in Fig. 6(d). We speculate that in the  $J$ - $V$  curves of nonpassivated devices, the nonradiative recombination current dominates, making it challenging to isolate the radiative recombination current from the overall current. A more systematic study is warranted to better understand the underlying mechanism at play.

#### IV. CONCLUSION

Our comprehensive studies unequivocally demonstrate the substantial enhancement of InGaN-based Micro-LED device performance through sidewall passivation with the  $\text{Al}_2\text{O}_3$  layer. Most notably, this  $\text{Al}_2\text{O}_3$  passivation layer significantly elevates the IQE of the Micro-LEDs, surpassing the 90% threshold. Remarkably, we achieved an IQE of 95.6% for passivated Micro-LED with a 10  $\mu\text{m}$  size, resulting in an EQE of 25.2% while maintaining an LEE of 26.3%. These findings provide invaluable insights into the potential avenues for further EQE improvements.

Practically, achieving an IQE of 100% for the real diodes operating at room temperature is a formidable challenge. Thus, attaining a 95.6% IQE represents a significant milestone and could indeed constitute the upper limit for Micro-LEDs.

Consequently, our pursuit of further improvements in EQE must focus on enhancing the LEE through innovative device designs.

In addition, our observations reveal a strong correlation between the diode ideality factor and IQE, where a lower ideality factor signifies a higher IQE. Our results emphasize that an ideality factor below 1.5 is a robust indicator of a high IQE exceeding 90%.

Finally, we have experimentally confirmed, for the first time, the unity ideality factor of the radiative recombination current in InGaN-based Micro-LEDs. This validation underscores the applicability of the SRH analysis to InGaN-based Micro-LEDs and contributes to our fundamental understanding of the underlying mechanisms governing their operation.

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